The phonological word: At the interface of lexical insertion, (morpho)syntax, and pronunciation

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1 Introduction

The PHONOLOGICAL WORD or PROSODIC WORD (pw), as the name suggests, is a phonological unit partly corresponding to the grammatical word, which is a pervasive and yet notoriously elusive entity. Bloomfield (1926:156) defines the (grammatical) word as a "minimum free form", by which he means that it is the smallest unit which can be used by speakers in an utterance. For instance, a speaker of English can say "Shark!" as a warning, or "Cookie?" as an offer, or "Four" as the answer to a question; hence *shark* and *cookie* and *four* are words by Bloomfield's definition.

This, of course, does not serve as a scientific definition of a grammatical word. Nevertheless, it is useful for our purposes since it anchors the notion of phonological word, which will be subject to whatever phonological constraints hold of a "minimum free form" in any given language, when spoken in the context of a longer utterance.

A widespread view today is that grammatical words can be identified in all languages (Dixon and Aikhenvald 2002:19, Dixon 2012:Vol. II: 7) on the basis of criteria such as cohesiveness, fixed ordering and conventionalised aspects of "coherence and meaning". A mostly complementary view defines them in terms of grammatical concepts such as roots and affixes (as in Haspelmath 2023).

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Dixon and Aikhenvald (2002:15) describe a phenomenon-based approach to identifying phonological words based on co-occurring segmental and prosodic properties and phonological rules. Schiering et al. (2010) argue that such clusters of phenomena may not end up identifying a unique phonological word at all, but instead point to the existence of any number of roughly word-sized units defined by partially conflicting criteria. In contrast to these two perspectives, we will sketch an approach to the phonological word based on Chomsky's (1986) notion of i-language, the speaker's computational system for language, and its interfaces. In brief, we will suggest that phonological words are the prosodic result of a procedure by which the phonological content of grammatical words is accessed, assembled and stored in short term memory preparatory to pronunciation. The procedure can be assumed to occur in all languages, even if the properties of the phonological word emerge differently in different languages.

In order to get one step closer to a scientific definition, we can assume that there is an underlying syntactic word (SW), and that the mapping from underlying syntactic words to surface phonological words adheres to certain universal principles, principles which allow a limited range of mismatches between the two. (The term 'grammatical word' is potentially broader and vaguer than the term 'syntactic word', since phonology is part of the grammar; we assume that the syntactic word can be defined independently of its phonological exponence, and use grammatical word only when we are being vaguer, as in our discussion of Bloomfield; see discussion of the syntactic word in Svenonius 2016b.)

The most common kinds of mismatch between the syntactic word and the phonological word are (i) when a single syntactic word is made up of two (or more) phonological words, as is the case with compounds in many languages, and (ii) when a single phonological word is made up of a syntactic word plus some additional clitic material.

The two cases are illustrated schematically in (1), following Dixon and Aikhenvald (2002) in considering clitics to be syntactic words, for the purpose of illustration.[1]

(1) a. compound: $[_{\mathrm{SW}}\ [_{\mathrm{pw}}$ table$][_{\mathrm{pw}}$ leg$]\ ]$
 b. word with clitic: $[_{\mathrm{pw}}\ [_{\mathrm{SW}}$ cat$][_{\mathrm{SW}}$'s$]\ ]$ (as in *The cat's here*)

The term CLITIC is used with a range of definitions, but the core meaning is that of a phonologically dependent syntactic word. A useful distinction among clitics is

[1]It is not uncontroversial that the tensed copula *is* is a grammatical word by Bloomfield's diagnostic, but in support of this possibility, imagine an argument between schoolchildren devolving into *Is! – Is not!*

expressed in terms introduced by Zwicky (1977): simple clitics are prosodically dependent words whose linear positioning is determined entirely by regular rules of syntax, while the position of special clitics is governed by a different rule from those applying to full phrases, typically requiring adjacency to a particular kind of host such as a finite verb. An example of a simple clitic is the English reduced copula, as in (1b), which is placed according to the usual rules of syntax. An example of a special clitic is the Spanish object pronoun *la* in *Yo la tengo* 'I have it', since it is left-adjacent to the verb, unlike the usual placement for objects in Spanish.

In the examples in (1), the relationship between pw and SW is one of containment. Sometimes it is argued that mismatches can exist in both directions at once. One type of mismatch may arise as a result of the kind of resyllabification that may occur when words are combined. For example, Peperkamp (1997:28f.) argues that in Italian *bar aperto* 'open bar', the phonological words are actually [ba] and [raperto], with resyllabification of /r/, so that the first syntactic word is split between the two phonological words, and the second phonological word includes a part of the first syntactic word.

Dixon (2020:26f.) suggests a similar kind of nonisomorphism for an example from Fijian (fij; Austronesian; Fiji). The example he discusses (p. 36) involves a functional element in Fijian which he analyzes as a nominaliser and part of a syntactic word with the verb with which it combines (he uses the term 'grammatical word'), illustrated in (2a). This nominaliser can become part of a phonological word together with a determiner, which is a distinct syntactic word, as illustrated in (2b), with bracketing on separate lines following Dixon's analysis in (2c).

(2) a. i-séle
NOMLZ-cut
'knife'

b. á-i séle
the-NOMLZ cut
'the knife'

c. $[_{pw}$ á -i] $[_{pw}$séle]
$[_{SW}$the] $[_{SW}$-NOMLZ cut]

Whether Dixon's interpretation of this case is accepted will depend on various details. The idiosyncratic, listed nature of *i-séle* 'knife' does not prove that it is a word; compare for example the idiomatic and listed *make out* meaning 'discern' (see Di Sciullo and Williams 1987 on the difference between listemes and words). Furthermore, even if *i-séle* is a word in (2a), it might not be one in (2b), just as Bloomfield (1933) argues that a compound like *blackbird* does not contain the syntactic word *black* — in other words, just because a string of segments is a syntactic word in some uses doesn't mean that same string is always a syntactic word, even when it is linked to the same content.

The organisation of the remainder of this paper is as follows. Section 2 reviews phonological evidence for the phonological word. Section 3 addresses notions of the syntactic word and its mapping to the phonological word, and suggests how a grammar might productively generate words. Section 4 considers whether the phonological word may allow recursion or if distinct word-like categories such as the prosodic stem or clitic group (or composite group) level should be recognized. Finally, Section 5 concludes with a summary.

For other overviews of the phonological word, see Revithiadou (2011), Hildebrandt (2014), and Tallman (2020).

2 Evidence for the phonological word

2.1 Stress assignment and word minimality

Perhaps the most salient characteristic of the phonological word is as a bounding domain for stress assignment. There are languages which apparently do not have word stress, but Hyman (2006) argues that in those that do, stress is obligatory in the phonological word; this means that no stress language has a set of stressless content words. This contrasts with the situation in some tone languages, where content words that lack an underlying tone specification are possible.

Furthermore, stress is bounded by the word (cf. Kaisse 2017). As noted by Selkirk (1996:197), a sequence of two or more syllables before a main word stress in English will necessarily bear a secondary stress if they are contained in the same word (as in *intèrpretátion*), but not if they are not. For example, in *from in the státion* [frəm ɪn ðə ˈsteɪʃn̩] or *so you can stáy* [sə jə kn̩ ˈsteɪ], the sequences of three function words can all remain unstressed, surfacing in their 'weak' forms, showing that metrical feet do not have to be built there.[2]

A related difference is that stress is typically culminative within the phonological word: each phonological word has a single *main*-stressed syllable, although there

[2]Depending on speech tempo, it is natural for such function word sequences to have internal rhythmic organisation through the insertion of 'strong' forms, which diagnoses incorporation as phonological words, e.g. (with strong forms underlined) [<u>ˌsəʊ</u> jə kn̩ ˈsteɪ] or [sə <u>ˌjuː</u> kn̩ ˈsteɪ]. The sensitivity to performance factors suggests an extrasyntactic dimension. We leave the internal prosodic organisation of such sequences to future research.

may be more than one secondarily stressed syllable per phonological word in languages with a distinction between main and secondary stress. There are exceptions to culminativity: in Nyulnyul (nyv; Nyulnyulan; Australia; McGregor 2011), for example, all stressed syllables of the word are reported to be "about equally prominent".

Tone, on the other hand, is not generally culminative. Here, too, there are exceptions. In Ambel, or Waigeo (wgo; Austronesian; Indonesia; Arnold 2018), phonological words may bear a high tone or be tonally unspecified. Only one syllable per phonological word may bear high tone, however.

The domain of stress can be quite large. Consider the examples in (3) from Central Alaskan Yup'ik (esu; Eskimo-Aleut; United States; Woodbury 1987). Iambic feet are built left to right, with a degenerate foot constructed on any remaining odd-numbered syllable. Where the incorporated noun at the left edge of the word is monosyllabic (3a), it is unstressed, and stress resumes on the first syllable of the morpheme immediately following, in this case the verb *ssu-* 'hunt', and every other syllable thereafter. With a disyllabic incorporated noun (3b), the stress falls on the second syllable of the noun, and skips the initial syllable of the verb. (Woodbury 2024:117 presents these examples without the word-final stresses; whether stress is manifested on the final syllable depends in part on whether there are enclitics.)

(3) a.

pi-	ssú:-	tu-	llí:ni-	lú:ní
thing-	hunt-	always-	apparently-	APP.3REFL

'S/he apparently always hunted [things].'

b.

mallú:-	ssu-	tú:-	lliní:-	luní
beached.whale-	hunt-	always-	apparently-	APP.3REFL

'S/he apparently always hunted beached whales.'

Thus, an affix like *-llini* 'apparently' can be stressed on either syllable depending on the syllable count of the string appearing to its left. The subparts of these utterances are inseparable except by other bound morphemes. Syntactic operations such as topicalization and question formation cannot reorder them. The subparts generally cannot be uttered in isolation, and so it is reasonable to regard the entire utterance as a single word, independently of the stress facts. (See Compton and Pittman 2010 and Arnhold et al. 2023 for evidence of similarly large words in the related language Inuktitut.)

Another property strongly associated with phonological wordhood is the minimal word requirement (Hayes 1995), which places a lower limit on phonological word size, usually disyllabic or heavy monosyllable (on syllable structure, see Côté, this volume).

English is an example in which the minimal word can be described as bimoraic: content words must be minimally $\sigma_{\mu\mu}$, where the second 'weak' mora is contributed by the coda, as in *cut*, or forms part of the nucleus, as in *key*. For the purposes of word size, $\sigma_{\mu\mu}$ is equivalent to $\sigma_{\mu}\sigma_{\mu}$ disyllables like *villa*, *feta*, or *comma*. Weak function words can appear as σ_{μ} (e.g., *the*) only as part of a larger utterance.

Not all languages impose a minimum word requirement greater than one syllable, or one mora. The Cois Fhairrge dialect of Irish (Ó Siadhail 1988), for example, has a contrast between long and short vowels, but permits monomoraic content words such as /tro/ 'foot' /u/ 'egg', and /tʹe/ 'warm' (where the prime mark indicates palatalization). If there is a minimum word requirement here, it is one monomoraic syllable.

An example of a language with a disyllabic minimum is North Saami (sme; Uralic; Norway; Sammallahti 1998). The minimal word size for content words is the disyllabic trochee, as seen in the adaptation of the Norwegian *bil* 'car' to the North Saami *biila* /pij:la/. This requirement does not apply to function words, however, including referring expressions like pronouns, e.g. *mun* /mu:n/ 'I'.

Vietnamese (vie; Austroasiatic; Vietnam) is an analytic language in which all morphemes are arguably monosyllabic. Certain names and loanwords might be exceptions, but there is evidence that even a loan like *cà-phê* 'coffee' is an idiom consisting of two cranberry morphs (Nhàn 1984). Schiering, Bickel, and Hildebrandt (2010) argue there is no evidence for the phonological word in Vietnamese. According to them, the language lacks a minimal word constraint, since CV words such as *đi* (/di/) 'go' are permitted. Brunelle (2015), however, reports that vowels lengthen obligatorily in open syllables, indicating that a bimoraic minimum is active in the language.[3] Although it is logically possible that the bimoraic minimum is a property of the Vietnamese syllable rather than the phonological word, this would not be a standard assumption of syllable theory. Pham (2008) further shows that some function words in Vietnamese can cliticise in a reduced phonological form. Such reduction may apply to *ở* when it is a preposition meaning 'at', but not to its lexical homophone *ở* [ʔəː˨˩], which means 'live, stay'. Since it is crosslinguistically typical that function words need not surface as phonological words, this suggests that Vietnamese has a bimoraic minimum at the level of the phonological word, and that function words like prepositions are exempt because they need not map to phonological words.

Hayes (1995) connects the minimal word constraint to constraints on foot size, suggesting that in some cases, the minimal word may follow from a requirement that a word must minimally contain a foot. (Ito and Mester 2008 suggest that it could follow from the headedness requirement of Prince 1980.) However, Garrett (1999)

[3]Thompson (1965) reports that /i/ in this context undergoes diphthongisation: [ʔɗɪi˧].

shows that there are at least some languages in which minimal word size cannot be reduced to foot well-formedness. In fact, he argues that there is a double dissociation: there are languages with a disyllabic minimum which nevertheless allow monosyllabic feet in longer words, and there are languages in which CVC satisfies the minimality requirement but patterns with CV as light for foot construction.

2.2 Phonotactics and phonological rules

The phonological word is the domain for a great number of phonological generalisations. We mention a few examples here. In Basque (eus; isolate; Spain, France), rhotics are prohibited from word-initial position (Hualde and de Urbina 2003). In Brokpa (sgt; Sino-Tibetan; Bhutan; Wangdi 2021), certain segments, including the glottal stop /ʔ/ and the voiceless sonorants /l̥ r̥/ only occur at the beginning of the word. In Bunaq (bfn; Timor-Alar-Pantar; Timor), only word-final syllables may be closed (Schapper 2022). In Germanic languages more generally, the right edge of the phonological word is often argued to license an APPENDIX which gives rise to clusters which are not found word-internally (as in *prompt*, where *t* is an appendix).

The phonological word is also the domain for rules affecting segments not situated at the edge. Italian (ita; Indo-European; Italy) has a rule of intervocalic *s*-voicing that applies within the phonological word (Nespor and Vogel 1986:124ff.), e.g. *a*[z]*ola* 'button hole', *di*[z]*attento* 'in-attentive', vs. *di*[s]*tratto* 'distracted'. The voicing rule does not apply between phonological words, e.g. *gatto* [s]*iamese*, *[z]*iamese* 'Siamese cat'.

Suprasegmental phenomena frequently have the phonological word as their domain. For example, in Cavineña (cav; Pano-Tacanan; Bolivia; Guillaume 2008), each prosodic word is mapped to a HMM tone contour. In words of three syllables or more, the H tone (marked with an acute accent) spreads as far as the antepenultimate syllable, and the penultimate and final syllables are assigned M tone (unmarked), e.g. *kímisha* 'three', *jútákiju* 'therefore', *wésá-tána-tsu* (lift-PASS-SS), *nétí-dírú-kware* (stand-GO.PERM-REM.PAST). In disyllabic words, H and M tone are assigned to the first and second syllable, e.g. *béta* 'two'.

2.3 The phonological word in phrasal context

Certain phonological rules, SANDHI rules, apply specifically at the boundaries between phonological words. A well known example of this is final voicing in Sanskrit (sa; Indo-European; South(east) Asia ca. 1500 BCE–1350 CE; Whitney 1889), which voices an obstruent before a voiced segment between words but not between morphemes of the same word (Selkirk 1980), e.g. /mahat/ ‘great’ + /ākhyānam/ ‘narrative’ > *mahad ākhyānam*, vs. /mahat-/ ‘great’ + /-ā/ ‘instr. sg.’ > *mahatā* (not **mahadā*; Kessler 1994). Similarly, sibilants in Catalan are voiced before a vowel across a word boundary (Wheeler 2005:162), e.g. *cas extrem* (*ca*[z]) ‘extreme case’, but retain their underlying voice specification before a vowel internal to a word (as in *gossa*, [ˈgo.sə] ‘(female) dog’). In both cases, the rule or constraint would be said to apply at the phrasal or post-lexical level, and to apply only to environments which are derived by the concatenation of words at that level.

Phrasal phonology also provides a different kind of evidence for the phonological word. Many languages exhibit a binarity requirement on the phonological phrase, whereby phonological phrases preferentially contain two prosodic words (Binarity(ϕ); Ito and Mester 2009). For example, in the sentence *She loaned her rollerblades to Robin* (originally from Selkirk 2000), the minor words *she*, *her* and *to* are assumed to procliticise to the following lexical word, resulting in three maximal phonological words (ω): (She loaned $)_\omega$ (her rollerblades $)_\omega$ (to Robin $)_\omega$. This may be parsed into a single phonological phrase (ϕ) or two, as shown in (4).

(4) a. { (She loaned $)_\omega$ (her rollerblades $)_\omega$ (to Robin $)_\omega$ $\}_\phi$
    b. { (She loaned $)_\omega$ (her rollerblades $)_\omega$ $\}_\phi$ { (to Robin $)_\omega$ $\}_\phi$

In (4a) we have a single phonological phrase made up of three phonological words, while the second phonological phrase of (4b) contains a single phonological word. The variability is ascribed to the fact that there is no way of satisfying Binarity(ϕ). Contrast this with the prosody of the syntactically equivalent *She loaned her rollerblades to Robin's sister* in (5).

(5) { (She loaned $)_\omega$ (her rollerblades $)_\omega$ $\}_\phi$ { (to Robin's $)_\omega$ (sister $)_\omega$ $\}_\phi$

Unlike (4), (5) only admits of a parse into two phonological phrases, each of them binary. A division into a ternary and a unitary phonological phrase as in (6) would be prosodically ill-formed since it would be harmonically bounded by (5). (See Prince and Smolensky 2004 for an explanation of this notion.)

(6) *{ (She loaned $)_\omega$ (her rollerblades $)_\omega$ (to Robin's $)_\omega$ $\}_\phi$ { (sister $)_\omega$ $\}_\phi$

2.4 The phonological word as emergent

We've discussed typical features of phonological words, including stress, minimality, and the function of the word as a domain for phonological rules. For many languages, the phonological word is crucial in comprehensive descriptions. However, questions remain about whether a single characterization applies across languages and if the phonological word is a universal concept. Mainstream phonology has tended to assume so, although the specific rules and constraints which refer to it may vary from language to language.

Schiering et al. (2010) reject the mainstream position, arguing that a prosodic domain which we call the word emerges in some languages as a domain for the phenomena described in sections 2.1 and 2.2, and fails to do so in others. Furthermore, in some languages, there is evidence for more than one phonological word-like domain.

Schiering et al. base their argument on two case studies, Vietnamese, already discussed in Section 2.1, and Limbu. For Limbu (lif; Sino-Tibetan; Sikkim, Nepal; Driem 1987), they argue that more than one prosodic level is needed between the foot and the phonological phrase, and that this proliferation of levels undermines the rationale for a universal phonological word. Since each domain is associated with an arbitrary cluster of phenomena, it suggests that the domains themselves are innovated on a language-particular basis.

We return to the question of the universality of the phonological word in Section 3.3.

3 Interpreting lexical and (morpho)syntactic words

3.1 Phonology as interpretive

Despite the kinds of mismatches mentioned above, prosodic word boundaries generally indicate syntactic word boundaries (with clitics being outside the domain for some diagnostics, such as stress in the examples mentioned in examples like *from in the státion* in subsection 2.1, and yet not forming distinct domains for other diagnostics, as in (4–5) in §2.3). The standard interpretation of this fact is that the phonological word is constructed on the basis of the syntactic word, which is either the output of a pre-syntactic Word Formation (WF) module or the output of syntax. It is within this domain that footing occurs, along with other 'lexical' phonological processes (in the sense of Lexical Phonology and Morphology, LPM). Phonological phrases, on the other hand, are based on larger constituents, including material not visible at the level of the word.

In either case, the building of (phonological) words on the output of WF or of syntax is an example of phonology being INTERPRETIVE, an idea which goes back at least to the era of the Standard Theory in syntax (Chomsky 1965). According to the classic statement of this view in *The Sound Pattern of English (SPE)* (Chomsky and Halle 1968), word boundaries (#) were inserted into syntactic surface structure prior to the phonological component. Word boundaries delimited the operation of the phonological rules that served to interpret the output of syntax.

The interpretive view is also at the basis of work on prosody beginning with Selkirk (1972). Non-linear approaches to prosody that emerged in the 1970s and 80s were based on the idea that phonologically interpreted sentences could be chunked into units organised in a PROSODIC HIERARCHY whose categories corresponded in some way with syntactic constituents (e.g. Nespor and Vogel 1986). (For the syntax-phonology interface, see Elfner, this volume.)

According to one widely current theory of the syntax-phonology interface (Match Theory; Selkirk 2011), three prosodic levels are identified as corresponding systematically to three syntactic levels, although with the possibility of mismatches between syntactic and prosodic constituency. Other things being equal, the clause maps to an intonational phrase (ι), the maximal XP to the prosodic phrase (ϕ), and the syntactic word to the phonological word (ω).

3.2 Lexical and syntactic word formation

Approaches to word formation are commonly broadly divided into two camps according to whether words are formed *prior* to their being combined into syntactic structures (Lexicalist) or *after* (Syntactic). Lexicalist approaches to word formation may be morpheme-based (e.g., Selkirk 1982, Kiparsky 1983, Williams 2007) or not (e.g., Kiparsky 1982a, Anderson 1992, Stump 2001), but are quite varied in several respects. Syntactic approaches also show great variation, differing on such matters as head movement, late insertion of exponents, the site for late insertion, and others, but the the most widely known contemporary representative of syntactic approaches is Distributed Morphology (DM, Halle and Marantz 1993). Syntactic approaches go back to Lieber (1980) and Sproat (1985), and also currently include Nanosyntax (Baunaz et al. 2018) and Spanning (Svenonius 2016b).

Syntactic approaches to word formation argue that the same mechanism which combines words into phrases and phrases into larger phrases (Merge, Chomsky 1995) is also responsible for combining morphemes into words. This is clearest in the case of Mirror Theory (Brody 2000) and Spanning Theory (Bye and Svenonius 2012, Svenonius 2016b), in which the complementation relation manifested by first Merge (external

Merge of a complement) is directly responsible for word formation: when T takes V as a complement, T becomes a suffix on V, unless something causes V to spell out separately from T.

In other theories, such as Koopman and Szabolcsi (2000), Julien (2002), and Nanosyntax, phrasal movement is centrally involved in word formation; movement is analyzed as a subcase of Merge (internal Merge). For example, VP might move to the specifier of T in order for T to spell out as a suffix on V. Ultimately, these approaches sometimes explicitly reject the notion of syntactic word (e.g., Julien 2002; 2007). Julien argues that whether an adjacent sequence of two heads spells out as a phonological word or not depends on their phonological properties, not their syntactic properties. In that case there is no word formation in the syntax, though syntax aligns the heads that spell out as phonological words. Clearly we do not adopt this position. Instead, we suggest that morphological and phonological differences among affixes, special clitics, simple clitics, and freestanding words can be derived from differences in their syntax.

In mainstream DM, movement is also central to word formation, but chiefly a special case of movement known as head movement, by which V can move to T, stranding its projections and dependents, to spell out with T as a suffix (Travis 1984, Baker 1988, Embick 2010).

The mainstream DM approach, in which head movement figures centrally, is the best known and most widely adopted. A syntactic word in this framework can be defined as a maximal X°, which is an X° dominated by no node with the zero bar-level designation marking it as a "head". Complex X° nodes are formed by head movement, which is generally licit when the target takes the maximal projection of the moved head as a complement.

3.3 Chunking (morpho)syntactic words and spans

Given a definition of the syntactic word (e.g., as a maximal X°), it is possible to define constraints that require this entity to map onto a phonological word, as in Match Theory. In derivational terms, at the spell-out stage of the derivation, the exponents which spell out the subparts of the syntactic word are retrieved from the lexicon and concatenated, and phonological processes, such as foot construction, apply (on cyclic word building, see Marvin 2002, Newell 2008, Shwayder 2015).

The output of each round of cyclic phonology is still a preliminary representation, and even the output of the final cycle is subject to further effects at the phrasal level (see section 4.3). Scheer (2011; 2012) argues that no independent notion of phonological word is needed; the word can simply be equated with the output of a cycle of spell-out, which Scheer hopes to derive from the syntactic PHASE. A challenge for this position is how to maintain the distinction between phonological words and phonological phrases, since the latter have also been argued to derive from syntactic phases (e.g., Adger 2007, Kratzer and Selkirk 2007, Kahnemuyipour 2009).

A functional motivation for the difference between words and phrases may come from properties that the word has but phrases and clauses do not. We will discuss two. The first property is that the word is the interface between the syntax and the lexicon. Accessing listed words and constructing novel words online both involve retrieval of stored items from a lexicon containing tens of thousands of items, and must take place several times per second at typical rates of speech.

A basic finding in psycholinguistics is that speech production proceeds in incremental stages, that is, the utterance is planned with subparts being prepared in parallel before articulation. Intonation breaks mark rounds of speech planning, including the selection of lexical items and the building of syntactic structure (Levelt 1989, Krivokapić 2007). Lexical access itself takes place in several stages, with the concept being accessed independently of its phonological form (Kempen and Huijbers 1983, Levelt 1989), and the underlying phonological form being accessed prior to the construction of the surface inflected form to be used (Caramazza et al. 1988, Levelt et al. 1999). Phonological representations are copied from long-term storage in the form of strings of phonemes, independently of prosodic information (Levelt 1989, Biran and Friedmann 2005), and stored short-term in a PHONOLOGICAL OUTPUT BUFFER (POB) in preparation for articulation (Friedmann et al. 2013, Dotan and Friedmann 2015). The buffer is limited in size (Butterworth 1992), generally accommodating single words, possibly with clitic material. Wheeldon and Lahiri (2002) and Wynne et al. (2018) argue from production experiments that compounds count as single words for the purposes of the buffer, as do function words when they are pronounced as phonological words (e.g., a non-cliticised pronoun in its 'strong' form), so that the operative unit for the phonological output buffer is a syntactic word which is also a phonological word—closely corresponding to the unit picked out by Bloomfield's operational definition of word.

It is possible that the properties most characteristic of the phonological word derive from how syntactic words are stored short-term, preparatory for pronunciation. Recall Hyman's observation that if a language has stress, all lexical words bear stress. This suggests that foot-building is something which happens automatically to satisfy the constraints of the phonological output buffer. The fact that the stress domain is always bounded by the phonological word suggests that foot-building cannot happen anywhere but the POB. The chunking of the word into feet may be an expression of deeper grouping principles that augment the effective storage capacity in working memory (cf. Miller 1956). In general, this grouping finds rhythmic expression in alternating (secondary) stress patterns, but the same principle may even be at work in the absence of audible secondary stress or other overt phonological indicators.

In tone languages whose roots are associated with a set of tonal melodies, the principles of autosegmental mapping may also be seen as a way of dividing the phonological word up into smaller tonally defined chunks (though the autosegmental nature of tone means that unlike stress, it can undergo postlexical reassociation across a word boundary in certain cases).

In Section 2.1, we noted Garrett's (1999) argument that word minimality is not necessarily tied to foot binarity. The double dissociation observed by Garrett suggests that the word minimum derives from some other factor than limitations on the POB. Function words may be exempt from the word minimum requirements that hold for content words (cf. Section 2.1 on North Saami), suggesting a connection to differences in the way content and function words are accessed and processed. Selecting a content word involves competition among semantically related lexical entries. A longer processing window for content words may therefore reduce the likelihood of errors in lexical choice. We speculate that the word minimum requirements that are manifested phonologically in many languages reflect the longer processing window.[4]

The second property that distinguishes words from phrases and clauses is morphology. Level- or stratum-ordering may offer another partial solution to efficient usage of working memory: pieces are accessed from the lexicon, admitted to the POB and processed incrementally. It may be no accident that the most convincing examples of cyclic rule application also involve stress assignment to successively build up morphological layers.

The POB provides a space in which items already admitted may exert an influence on which other items are retrieved from the lexicon. Not all languages exhibit the peculiarly morphological phenomena of declension and conjugation classes, suppletive allomorphy, portmanteaux, and irregular forms, but in those that do, such phenomena are usually restricted to the word. Even in languages with highly regular morphology, affixes can usually be distinguished from non-affixal light function words in the selectivity they show for their hosts; affixes are normally rigidly bound to the extended projection in which they are interpreted, something which is captured by the Head Movement Convention (Travis 1984) and the Mirror-Theoretic definition of spans in terms of complementation (Brody 2000, Svenonius 2016a).

[4]An alternative explanation might see the word minimum as a listener-oriented strategy for supporting fast and accurate recognition of content words, but we abstract away from this possibility given that it is not rooted in constraints on speech planning.

If the POB is where words are assembled and stored short term before they are prepared for pronunciation, then in order to be rapidly accessed by the articulatory mechanisms, they must appear in the buffer with their various morphological pieces, not simply as bare roots, stems or paradigms. This would be consistent with fully formed words being inserted into the buffer. However, studies of aphasia have found that deficits in the output buffer lead both to phonological errors for content word roots (e.g., *bayit* 'house' may be pronounced as the nonword *gayit*, with a phonological substitution, in Hebrew) and to substitutions of functional exponents with other exponents in the same class (e.g., dative *le-bayit* 'to the house' for non-dative *ha-bayit* 'the house', using an existing but incorrect form of the clitic determiner; Friedmann et al. 2013, Dotan and Friedmann 2015).

The fact that both kinds of error are regularly observed in a class of patients suggests a single locus for assembling the two kinds of information, namely the POB. There, phonological roots of content words are combined with high frequency functional exponents, and a single type of deficit can lead both to phonological errors in roots and to errors in the selection of the right form of a functional element (cf. Bye and Svenonius 2012 for a model of lexical insertion in which an array of functional exponents are activated for possible insertion before the correct one is selected).

We hypothesize, therefore, that the properties of the phonological word are the result of being processed in a short-term phonological output buffer (the stage that Levelt 1989 calls spell-out) before being accessed for phrase-level linearisation and articulation (as in such works as Shallice et al. 2000 and Dotan and Friedmann 2015). The POB takes in the parts of a word, as determined by the syntax, and creates a phonological word from it, including the construction of feet if the language has stress and the cyclic application of phonology.

The selection of lexical allomorphs is also determined on the basis of information available within a word domain, and we suggest that this is because the correct allomorph must be selected in the POB in preparation for spell-out. At least some special clitics must be present in the POB, namely those which are subject to lexical allomorphy or word-level phonology (including stress; the POB can distinguish elements inside and outside the stress domain, since some affixes are stress-neutral).

Once assembled and processed in the POB, the stress pattern and other properties that derive from the constraints of short-term storage cannot be recomputed postlexically (cf. the Strict Cycle Condition of Mascaró 1976 and related proposals). Any changes at the level of the phonological or intonational phrase should therefore not affect foot structure. As far as we know, this expectation is borne out: there are no languages in which stress is recomputed over a larger phrasal domain. (Recall that we conjectured that there is no source for prosodic feet outside the POB.)

An apparent counterexample to this claim would be the original analysis of the English RHYTHM RULE by Liberman and Prince (1977), since it assumes that expansion to a phrasal context can trigger adjustments in the grid representation of a word built in an earlier cycle. For example, *Japanese* has final main stress in isolation, but main stress apparently shifts to the initial syllable in the phrase *Japanese chairs*. In grid theory, this can be represented as the displacement of a level 3 grid mark from the final syllable of *Japanese* to the initial syllable, as shown in (7).

```
(7)                x                  x
            x      x          x       x
        x   x      x          x   x   x
        x x x      x          x x x   x
        Japanese chairs → Japanese chairs.
```

A reanalysis by Gussenhoven (1991), however, takes a step towards understanding the rhythm rule in entirely intonational terms. For Gussenhoven, accent is a placeholder for pitch accent, which is assigned post-lexically to accented syllables, either lexically marked or assigned by (lexical) rule. In the case of $\overset{*}{J}$*apan*$\overset{*}{e}$*se*, the final syllable is lexically accented; the initial syllable is accented by rule. The word forms a minimal pair with $\overset{*}{J}$*apanise*, which is identical in terms of foot structure, but lacks the lexical accent on the final syllable. Where $\overset{*}{J}$*apan*$\overset{*}{e}$*se* is the sole content of the phonological phrase, a H* pitch accent is assigned to the initial accent, and a H*L to the final. He then reformulates the rhythm rule as the deletion of an accent between two accents within the same phonological phrase: $\overset{*}{J}$*apan*$\overset{*}{e}$*se ch*$\overset{*}{a}$*irs* → $\overset{*}{J}$*apanese ch*$\overset{*}{a}$*irs*. Deletion of the medial accent bleeds later insertion of a pitch accent in this position. In this account, the accent deletion rule still involves destructive alteration post-lexically of lexically assigned accent, but it points the way to an account based on pitch accent alone. Such an account is outlined in Gussenhoven (2004:278), who proposes that the phonological phrase in English ideally requires pitch accents at both edges, the effect of the two Alignment constraints ALIGN(ϕ,T*,Right) and ALIGN(ϕ,T*,Left). Understood this way, the rhythm rule becomes an epiphenomenon of pitch accent distribution within the phonological phrase, making literal accent deletion unnecessary.

4 Strict layering or recursion?

As mentioned above, certain languages afford evidence of word-like phonological domains of different sizes. The larger word-like domains include compound words and clitic groups; the smaller includes what some scholars term the 'prosodic stem'. Given our arguments for a universal basis for the word, we must consider the nature of such categories' relation to the phonological word proper: should they be viewed as authentic units of prosody in their own right, or are they derived from the phonological word by recursion or adjunction?

4.1 Recursion

Recursion at the word level has long been assumed in morphology. For example, for the word *established*, SPE has the recursive representation $[_V[_V$establish$]$ PAST$]$ (Chomsky and Halle 1968:8). If such structures are mapped isomorphically to phonological representations, recursive words are produced. It is commonly assumed in LPM analyses of English (e.g. Halle and Mohanan 1985) that inflectional suffixes are added to a base which is already morphophonologically a word, in other words they derive words from words. See Booij and Lieber (1993) for further examples.

Whether phonology *per se* admits of recursion has been debated in the context of just how far the parallels between syntax and phonology extend. Once the notion of prosodic constituency is admitted, recursion becomes in principle possible. In actuality, though, this recursive potential is not realised in full. In contrast to syntax, in which a non-clausal phrase such as VP may contain a clausal CP, a prosodic category of level n is thought never to be dominated by a lower category $n - 1$, so that for example a phonological phrase cannot contain an intonational phrase. Early work in prosodic theory therefore adopted some version of the Strict Layering Hypothesis (SLH, Selkirk 1981; 1984, Nespor and Vogel 1986), which imposes a strict hierarchy on prosodic structures.

The strict hierarchy is compatible with a linear representation in terms of boundary markers (cf. for example Selkirk 1986), while the layered representations of syntax require an additional dimension (for example a distinction between left and right brackets).

In its original formulation, the SLH also forbade adjunction of one category to another, i.e. structures where some prosodic constituent of category n could dominate another of the same level n. This is again in contrast to syntax in which recursion of categories through adjunction is commonly motivated by distributional equivalence (since adjoining A to B results in B′ with the same distribution as B). Some researchers, such as Pinker and Jackendoff (2005a;b) and Neeleman and van de Koot (2006) argue, given the interpretive role of phonology, that it is too impoverished to manifest recursion of any kind.

However, there have been many arguments that prosodic constituents can be adjoined to prosodic constituents at the same level. Gussenhoven (2004:275–94, 315f.)

discusses examples at the level of both the phonological and intonation phrase. Selkirk (1996) proposes decomposing Strict Layering into four distinct constraints, including a violable nonrecursivity constraint. Violations of nonrecursivity allow limited prosodic recursion in the form of adjunction, while violations of an exhaustivity constraint allow the skipping of levels in the prosodic hierarchy.

Sometimes it is suggested that adjunction exists in phonology only as a vestige of the mapping from the syntax (e.g., Kabak and Revithiadou 2009), but others such as van der Hulst (2010) argue for adjunction even at lower prosodic levels, implying that phonology itself is a source of recursive structure.

The question of phonological recursion also arises in the division of labour between representation and procedure in analyses. Cyclic rule application may alter or destroy foot structure built by earlier cycles, so a representation after the application of a recursive rule may fail to explicitly encode recursion. Classic OT, however, is based on a single mapping of input to output, requiring that phonological processes apply simultaneously and are evaluated in parallel. In that framework, any word boundaries relevant to the computation of stress must be present in the output representation. In this spirit there are recursive word structures in prosodic representations in Selkirk (1996) for pronominal objects in English, Peperkamp (1997) for some derivational morphology, Ito and Mester (2006) for Japanese compounds, and Ito and Mester (2008) for various English function words, for example.

4.2 Prosodic structure below the phonological word

Detailed studies of morphophonology often argue for a category of stem, smaller than the word but also circumscribed by morphology or morphosyntax (e.g., Kiparsky 1985, Aronoff 1994). A recurring pattern is one in which suffixes are more tightly integrated with the root than are prefixes, so that the root plus suffixes can be identified as a unit excluding the prefixes, at the same time as other diagnostics show that the prefixes are contained in the word. Pesetsky's (1979) analysis of Russian prefixed verbs, which deeply influenced Lexical Phonology and Morphology, is a case in point. Although the semantics of a prefixed verb like *pod-žéčʹ* 'to set on fire' points to the prefix and verb root forming a unit excluding inflection, Pesetsky argues that the morphological and phonological evidence shows that the root together with its suffixes in fact form a domain distinct from the prefix (the form *žéčʹ* is derived by the fusion of a root /žĭg/ with an infinitival suffix /-tĭ/, where /ĭ/ is the front *yer*, a kind of vowel that alternates with zero, as discussed in Rubach 1985).

Facts like these have also been understood in representational terms through the addition of a 'prosodic stem' category in the prosodic hierarchy. For example, Downing and Kadenge (2020) argue that phonological processes in Shona (sna; Atlantic-Congo; Zimbabwe) can target a stem like *pá* in *ku-pá* 'to give' (where *ku-* marks the infinitive), though the monosyllabic *pá* falls below the bisyllabic minimal word requirement for Shona. The prefix must therefore count towards satisfying the word minimum, indicating that it is part of the word. The stem edge, on the other hand, is preserved in the resolution of hiatus. In Shona, hiatus is normally resolved within words by deletion of a vowel or by changing the first vowel to a glide, e.g., *tú-á-fámba* → *twáfamba* (they-TAM-walk) 'they have walked' (TAM = Tense-Aspect-Mood). However, if the hiatus straddles the boundary between a prefix and a prosodic stem, then both vowels are preserved through insertion of a glide in between: *ku-ímbá* → *kujímbá* (INF-sing) 'to sing' (Downing and Kadenge 2020:447). Neither of these hiatus resolution strategies is applied between words, however, pointing to a need to distinguish between phrasal, phonological word and prosodic stem domains.

The existence of such an intermediate category might be motivated by the primacy of the left edge of the root in processing, which has been invoked to explain other asymmetries between prefixes and suffixes (Hawkins and Gilligan 1988).

There are also various ways in which stem-level effects might emerge in some languages without according root+suffix domains a separate status in the prosodic hierarchy, which we have argued might be functionally underpinned by the need for a phonological output buffer in articulation planning.

4.3 Between the phonological word and the phrase

In addition to distinguishing between phonological words and phonological phrases, Nespor and Vogel (1986) and Hayes (1989) posit an intermediate level called the CLITIC GROUP (CG), which, according to Hayes (1989:207) can be defined "roughly as a single content word together with all contiguous grammatical words in the same syntactic constituent".

Similarly, Vogel (2009; 2020) proposes to unify the clitic group and the compound in a category that she terms the COMPOSITE GROUP (κ). One of the empirical motivations for a category on this level comes from intervocalic s-voicing in Italian. Although the process applies at the boundary between affix and stem (see Section 2.2), it does not apply to clitics, e.g., *comprando=*[s]*e=lo* 'buying it for oneself', or between the members of a compound, e.g., *porta* [s]*apone* 'soap dish'. There are also non-cohering prefixes that must be incorporated at the level of the composite group according to Vogel, e.g., *ri*[s]*alire* 'to go up again', cf. cohering (phonological word-internal) *ri*[z]*alire* 'to date back to'.

In a similar vein, Vigario (2010) proposes that compounds form a constituent between the phonological word and phonological phrase, which she calls the PROSODIC WORD GROUP (cf. Kabak and Revithiadou 2009).

We suggested in section 3.3 that special clitics are introduced in the POB, and mentioned evidence that compounds are constructed there as well. Certainly this is necessary when clitics or compounds are included in the stress domain, as for the Lucanian clitics analyzed by Peperkamp (1997).

A case of compounds which would have to pass through the POB for the same reason would be those in Greek (Nespor and Vogel 1986), where the entire compound is a single domain for stress. The stress may fall on a syllable not stressed in either of the components, e.g., *kúkla* 'doll' + *spíti* 'house' = *kuklóspito* 'dollhouse'. Kabardian (kbd; Abkhaz-Adyge; Russian Federation; Gordon and Applebaum 2010) is another example in which the compound word forms a stress domain on a par with the simplex word.

In other cases, special clitics and compounds show more prosodic independence but we take this to suggest that the POB can support layered word structures.

Consider for example Warlpiri (wbp; Pama-Nyungan; Australia), for which Pentland and Laughren (2004) propose hierarchised word domains, distinguishing an inner domain for the application of vowel harmony and an outer domain for the assignment of compound stress. In the compound verb /pirri+kiji+rnu/ 'scatter-throw-PAST', /kiji+rnu/ forms a domain within which regressive vowel harmony applies, giving [kuju-rnu]. The initial preverb /pirri/ is outside the domain for vowel harmony. The preverb and the main verb constitute separate domains for stress assignment, but the compound as a whole receives main stress on the preverb: ((ˈpirri-) (ˌkuju-rnu)). Schiering et al. (2010) discuss a case from Limbu which is similar in that it is the smaller domain that limits the segmental rule, while stress is assigned within the larger. If these examples involve the stress system implicated in the POB, then all of these larger compoundlike words must be constructed there.

Given a device like the phonological output buffer, or POB, which we have adopted from the psycholinguistic literature, the above evidence for clitic groups, composite groups, or prosodic word groups suggests that the POB is capable of maintaining some internal organization to a complex word across cycles, to the extent that it can differentiate the attachment of a clitic or incorporee from the attachment of an affix. All of the above cases might be treated in terms of word adjunction in the POB (essentially as proposed by Ito and Mester 2008).

However, there are also simple clitics which cannot be introduced in the POB. We have already mentioned the reduced copula and auxiliary of English as an example. It cannot be introduced in a POB of its own because it is unstressed. It cannot be introduced in the POB of its host because it does not form a syntactic constituent or span with its host (e.g., in [*the cat*]*'s here*, cf. (1b), the host is in the right branch of a specifier and could not form a syntactic word with the copula).

In fact, the majority of light function words in English (such as prepositions, determiners, and auxiliaries) do not show any signs of being special clitics and can be assumed to be simple clitics when they are not phonological words. As simple clitics they are introduced in the phrasal phonology, not the POB. As a result, they cannot be included in the stress domain of their host. They furthermore cannot condition lexical allomorphy in their neighbors, or show lexical allomorphy conditioned by adjacent words. The allomorphy of the English indefinite article *a*∼*an* is conditioned by an adjacent word, but it is phonological conditioning, not lexical conditioning (cf. e.g., Bonet 1991 on allomorphy of clitics which depends on syntactic features, so clearly not postlexical, and Bonet and Torres-Tamarit (2010) on lexical allomorphy in the host of the clitic, where different verbs have different lexically specified augments in the presence of a clitic, so also clearly not postlexical).

This means that what has been called the clitic group is not a unified phenomenon. This conclusion is further motivated by languages in which simple clitics are phonologically less integrated than special clitics, for example Halkomelem (Salish) as described by Gerdts and Werle (2014). In the phrasal phonology, it is possible that simple clitics are adjoined to the prosodic word which is the output of the POB, but the phonological processes that affect them are postlexical and may differ from the phonology that affects special clitics introduced in the POB.

One of those postlexical processes, discussed by Selkirk (1972:191) and which may be cited as evidence for the existence of the clitic group at the phrasal level, is the deletion of /v/—possibly a gradient gesture reduction along the lines of Articulatory Phonology (Browman and Goldstein 1992). (See Gafos, this volume.) A word-final /v/ may delete preceding a clitic in the same clitic group, as shown in (8a). The same process does not occur finally within the clitic group, however, as shown in (8b). We thus find /v/-deletion preceding a cliticised (*some*), but not before a full NP object (e.g. *six*).

(8) a. [Save some]$_{CG}$ [for me]$_{CG}$, [will you]$_{CG}$?
[seɪ]

b. [Save]$_{CG}$ [six]$_{CG}$ [for me]$_{CG}$, [will you]$_{CG}$?
[seɪv], *[seɪ]

This supports the existence of a constituent like the clitic group at the phrasal level. It is possible that it can be analyzed in terms of prosodic word adjunction, like the adjunction structures formed earlier in the POB. Nevertheless, since these later adjuncts are not subject to the lexical phonology but only to the distinct postlexical phonology, it may be more perspicuous to refer to the constituent by a different term such as clitic group (as in (8)). This would apply to the constituents which were marked as words in (4–5) following Selkirk (2000) and Ito and Mester (2009). We think the motivation for the two levels of word spell-out and phrasal spell-out ameliorates Ito and Mester's motivations for keeping the prosodic inventory to a minimum; since the structures are spelled out in two distinct stages, word and phrase, there is potentially external motivation for two kinds of augmentation of the prosodic word, word-level augmentation leading to word adjunction, and phrase-level augmentation, which might lead to something different.

The possibility of prosodic adjunction of word-sized elements at the phrasal levels opens up the possibilty of a difference between compounds formed in the POB and phrasal compounds formed from distinct POBs composed as such in the phrasal component; there is no space here to pursue a discussion of phrasal compounds, on which see Trips and Kornfilt (2017) and references there.

5 Conclusions

The phonological word is not accepted by all scholars as a universal category of the prosodic hierarchy. We have suggested, however, that there may be a useful sense in which it is a universal category. Within the interpretive view of phonology that originates in *SPE* (Chomsky and Halle 1968), externalizing the syntactic representation entails retrieving and assembling the phonological roots, affixes, and other morphemes from the lexicon, making phonological adjustments to these structures in one or more cycles, and storing the resulting information short-term in a phonological output buffer (POB) before articulating it within larger phonological and intonational phrases. We have suggested that phonological word phenomena may be the prosodic expression of this buffer and, as such, the inevitable result of the three-way interaction of syntax, the selection of lexical entries and the articulatory-perceptual interface. Every language will have a phonological object which is the result of at least one cycle of lexical insertion into the POB. In addition, every language will have a phonological object which is copied from the POB into a plan for phonetically realising larger utterance chunks such as the phonological and intonational phrase.

We have conjectured that short-term memory limitations on the POB may be the source of two of the phenomena most characteristic of phonological words. The first is that the word may be chunked into smaller units through the grouping of syllables into metrical feet, although nothing requires a language to have stress, or any other specific phonological manifestation of the word. The second is that morphemes may be admitted to the buffer and phonological adjustments applied to their combinations in more than just one cycle. Although not all phonological processes in word formation are cyclic, it appears that cyclic rules are absent from the higher prosodic levels (an observation which was much discussed in the early days of lexical phonology; cf. Kiparsky 1982b, Kaisse and Shaw 1985, or Kaisse and McMahon 2011 and Scheer 2011 for more recent discussion).

We have not discussed in any detail the nature of the objects which feed into the POB, the syntactic words, but on standard assumptions they would correspond either to complex heads or to spans in the syntax. This implies a difference in whether and how special and simple clitics are integrated within the prosodic word.

At least some special clitics (such as the object clitics of Lucanian) are treated on a par with affixes in that they are fully included in the stress domain. Other special clitics, which are stress-neutral, fall outside the stress domain, but are nonetheless part of the phonological word input to the phrasal phonology, and so must be introduced in the POB after stress has been assigned.

On the other hand, we can assume that simple clitics like the English copula *'s* (in *The cat's here*) are excluded from the POB, and can only be attached in the phrasal phonology (this conclusion is forced on the assumption that the copula cannot syntactically form a complex head or a span with a subpart of the subject noun phrase). This predicts that such simple clitics will not be integrated into the stress domains of their hosts, count towards satisfaction of the minimum word requirement, or show cyclic phonological effects (on possessive *'s*, cf. Zwicky 1987).

If the sequence *cat's* is neither formed in the POB nor copied from the POB to the larger utterance plan, it cannot given our assumptions to this point be a phonological word in either of the senses above. This may be the right approach to take, although it would mean that the obligatory voicing assimilation seen there (and epenthesis between sibilants) can be postlexical.

It may however be desirable to recognize a phonological word which persists after alterations in the phrasal phonology. Postlexical resyllabification of the kind discussed by Peperkamp (1997) would be an example of this kind of constrained adjustment (recall *bar aperto* → $(\text{ba})_\omega$ $(\text{raperto})_\omega$ from section 1). If resyllabification can alter phonological word affiliation, resulting in a 'postlexical' or 'surface' word, late incorporation of simple clitics may be within the power of phrasal phonology as well. (For recent evidence for the reality of resyllabification, see Beristain 2024.) Whether this postlexical object should be called a word might be a terminological question.

References

Adger, David. 2007. Stress and phasal syntax. *Linguistic Analysis* 33:238–266.

Anderson, Stephen R. 1992. *A-Morphous Morphology*. Cambridge: Cambridge University Press.

Arnhold, Anja, Emily Elena, and Richard Compton. 2023. Inuktitut and the concept of word-level prominence. In *Word prominence in languages with complex morphology*, ed. Ksenia Bogomolets and Harry van der Hulst, 138–177. Oxford: Oxford University Press.

Arnold, Laura. 2018. A grammar of Ambel: an Austronesian language of Raja Ampat, west New Guinea. Doctoral Dissertation, University of Edinburgh.

Aronoff, Mark. 1994. *Morphology by Itself*. Cambridge, MA: The MIT Press.

Baker, Mark C. 1988. *Incorporation: A theory of grammatical function changing*. Chicago: University of Chicago Press.

Baunaz, Lena, Karen De Clercq, Liliane Haegeman, and Eric Lander, ed. 2018. *Exploring Nanosyntax*. Oxford: Oxford University Press.

Beristain, Ander. 2024. The role of (re)syllabification on coarticulatory nasalisation: aerodynamic evidence from Spanish. *Languages* 9:219.

Biran, Michal, and Naama Friedmann. 2005. From phonological paraphasias to the structure of the phonological output lexicon. *Language and Cognitive processes* 20:589–616.

Bloomfield, Leonard. 1926. A set of postulates for the science of language. *Language* 2:153–164.

Bloomfield, Leonard. 1933. *Language*. New York: Holt Rinehart and Winston.

Bonet, Eulalia. 1991. Morphology after syntax: Pronominal clitics in Romance. Doctoral Dissertation, MIT, Cambridge, Ma.

Bonet, Eulàlia, and Francesc Torres-Tamarit. 2010. Allomorphy in pre-clitic imperatives in formenteran catalan. In *Romance Linguistics 2009*, ed. Sonia Colina, Antxon Olarrea, and Ana Maria Carvalho, 337–352. Amsterdam: John Benjamins.

Booij, Geert, and Rochelle Lieber. 1993. On the simultaneity of morphological and prosodic structure. In *Studies in Lexical Phonology*, ed. Sharon Hargus and Ellen M. Kaisse, 23–44. San Diego, CA: Academic Press.

Brody, Michael. 2000. Mirror Theory: Syntactic representation in Perfect Syntax. *Linguistic Inquiry* 31:29–56.

Browman, Catherine P., and Louis Goldstein. 1992. Articulatory Phonology: an overview. *Phonetica* 49:155–180.

Brunelle, Marc. 2015. Vietnamese (Tiêng Viêt). In *The handbook of Austroasiatic languages*, ed. Mathias Jenny and Paul Sidwell, volume 2, 909–953. Leiden: Brill.

Butterworth, Brian. 1992. Disorders of phonological encoding. *Cognition* 42:261–286.

Bye, Patrik, and Peter Svenonius. 2012. Exponence, phonology and non-concatenative morphology. In *The morphology and phonology of exponence: the state of the art*, ed. Jochen Trommer, 427–495. Oxford: Oxford University Press.

Caramazza, Alfonso, Alessandro Laudanna, and Cristina Romani. 1988. Lexical access and inflectional morphology. *Cognition* 28:297–332.

Chomsky, Noam. 1965. *Aspects of the theory of syntax*. Cambridge, MA: MIT Press.

Chomsky, Noam. 1986. *Knowledge of Language*. New York: Praeger.

Chomsky, Noam. 1995. *The minimalist program*. Cambridge, Ma.: MIT Press.

Chomsky, Noam, and Morris Halle. 1968. *The sound pattern of English*. Cambridge, MA: MIT Press.

Compton, Richard, and Christine Pittman. 2010. Word-formation by phase in Inuit. *Lingua* 120:2167–2192.

Di Sciullo, Anna Maria, and Edwin Williams. 1987. *On the Definition of Word*. Cambridge, MA: The MIT Press.

Dixon, R. M. W. 2012. *Basic linguistic theory. Vol. 1: Methodology; Vol. 2: Grammatical topics*. Oxford: Oxford University Press.

Dixon, R. M. W. 2020. Words within words: examples from Yidiñ, Jarawara, and Fijian. In *Phonological word and grammatical word: a cross-linguistic typology*, ed. Alexandra Y. Aikhenvald, R. M. W. Dixon, and Nathan M. White, 25–38. Oxford University Press.

Dixon, R. M. W., and Alexandra Y. Aikhenvald, ed. 2002. *Word. A cross-linguistic typology*. Cambridge: Cambridge University Press.

Dotan, Dror, and Naama Friedmann. 2015. Steps towards understanding the phonological output buffer and its role in the production of numbers, morphemes, and function words. *Cortex* 63:317–351.

Downing, Laura J., and Maxwell Kadenge. 2020. Re-placing PStem in the prosodic hierarchy. *The Linguistic Review* 37:433–461.

Driem, Georg van. 1987. *A grammar of Limbu*. Berlin: Mouton de Gruyter.

Embick, David. 2010. *Localism versus globalism in morphology and phonology*. Cambridge, Ma.: MIT Press.

Friedmann, Naama, Michal Biran, and Dror Dotan. 2013. Lexical retrieval and its breakdown in aphasia and developmental language impairment. In *The cambridge handbook of biolinguistics*, ed. Cedric Boeckx and Kleanthes K. Grohmann, 350–374. Cambridge: Cambridge University Press.

Garrett, Edward. 1999. Minimal words aren't minimal feet. *UCLA Working Papers in Linguistics* 1:68–105.

Gerdts, Donna B., and Adam Werle. 2014. Halkomelem clitic types. *Morphology*

24:245–281.

Gordon, Matthew, and Ayla Applebaum. 2010. Prosodic fusion and minimality in Kabardian. *Phonology* 27:45–76.

Guillaume, Antoine. 2008. *A grammar of Cavineña*. Berlin: Mouton de Gruyter.

Gussenhoven, Carlos. 1991. The English Rhythm Rule as an accent deletion rule. *Phonology* 8:1–35.

Gussenhoven, Carlos. 2004. *The phonology of tone and intonation*. Cambridge: Cambridge University Press.

Halle, Morris, and Alec Marantz. 1993. Distributed Morphology and the pieces of inflection. In *The View from Building 20: Essays in Honor of Sylvain Bromberger*, ed. Kenneth Hale and Samuel J. Keyser, 111–176. Cambridge, MA: The MIT Press.

Halle, Morris, and K. P. Mohanan. 1985. Segmental phonology of Modern English. *Linguistic Inquiry* 16:57–116.

Haspelmath, Martin. 2023. Types of clitics in the world's languages. *Linguistic Typology at the Crossroads* 3:1–59.

Hawkins, John A., and Gary Gilligan. 1988. Prefixing and suffixing universals in relation to basic word order. *Lingua* 74:219–259.

Hayes, Bruce. 1989. The prosodic hierarchy in meter. In *Phonetics and phonology, Volume 1: Rhythm and meter*, ed. Paul Kiparsky and Gilbert Youmans, 201–260. San Diego, CA: Academic Press.

Hayes, Bruce. 1995. *Metrical Stress Theory: Principles and Case Studies*. Chicago: Chicago University Press.

Hildebrandt, Kristine A. 2014. The prosodic word. In *The Oxford handbook of the word*, ed. John R. Taylor. Oxford: Oxford University Press.

Hualde, José Ignacio, and Jon Ortiz de Urbina. 2003. *A grammar of Basque*, volume 26 of *Mouton Grammar Library*. Berlin: Mouton de Gruyter.

Hulst, Harry van der. 2010. A note on recursion in phonology. In *Recursion and human language*, ed. Harry van der Hulst, 301–341. Berlin: Walter de Gruyter.

Hyman, Larry M. 2006. Word-prosodic typology. *Phonology* 23:225–257.

Ito, Junko, and Armin Mester. 2006. Prosodic adjunction in Japanese compounds. In *Proceedings of The 4th Formal Approaches to Japanese Linguistics Conference (FAJL 4)*, volume 55 of *MIT Working Papers in Linguistics*, 97–112. Cambridge, MA: MIT Department of Linguistics and Philosophy.

Ito, Junko, and Armin Mester. 2008. The onset of the prosodic word. In *Phonological argumentation: essays on evidence and motivation*, ed. Steve Parker. London: Equinox.

Ito, Junko, and Armin Mester. 2009. The extended prosodic word. In *Phonological domains: universals and deviations*, ed. Bariş Kabak and Janet Grijzenhout, 135–194.

Berlin: Mouton de Gruyter.
Julien, Marit. 2002. *Syntactic Heads and Word Formation*. Oxford: Oxford University Press.
Julien, Marit. 2007. On the relation between morphology and syntax. In *The Oxford Handbook of Linguistic Interfaces*, ed. Gillian Ramchand and Charles Reiss, 209–238. Oxford: Oxford University Press.
Kabak, Barış, and Anthi Revithiadou. 2009. An interface approach to prosodic word recursion. In *Phonological domains: universals and deviations*, ed. Janet Grijzenhout and Barış Kabak, 15–46. Berlin: Mouton de Gruyter.
Kahnemuyipour, Arsalan. 2009. *The syntax of sentential stress*. Oxford: Oxford University Press.
Kaisse, Ellen. 2017. The domain of stress assignment: Word-boundedness and frequent collocation. In *On looking into words (and beyond)*, ed. Claire Bowern, Laurence Horn, and Raffaella Zanuttini, 17–41. Berlin: Language Science Press.
Kaisse, Ellen M., and April McMahon. 2011. Lexical Phonology and the lexical syndrome. In *The Blackwell companion to phonology. Volume IV: Phonological interfaces*, ed. Marc van Oostendorp, Colin J. Ewen, Elizabeth Hume, and Keren Rice. Malden, MA: Wiley-Blackwell.
Kaisse, Ellen M., and Patricia A. Shaw. 1985. On the theory of lexical phonology. *Phonology Yearbook* 2:1–30.
Kempen, Gerard, and Pieter Huijbers. 1983. The lexicalization process in sentence production and naming: Indirect election of words. *Cognition* 14:185–209.
Kessler, Brett. 1994. Sandhi and syllables in Classical Sanskrit. In *The proceedings of the Twelfth West Coast Conference on Formal Linguistics*, ed. Erin Duncan, Donka Farkas, and Philip Spaelti, 35–50. Stanford, Ca.: CSLI.
Kiparsky, Paul. 1982a. Lexical Morphology and Phonology. In *Linguistics in the morning calm. Selected papers from SICOL-1981*, ed. Linguistic Society of Korea, 3–91. Seoul: Hanshin.
Kiparsky, Paul. 1982b. From Cyclic to Lexical Phonology. In *The structure of phonological representations, Vol. 1*, ed. Harry van der Hulst and Norval Smith, 131–175. Dordrecht: Foris.
Kiparsky, Paul. 1983. Word formation and the lexicon. In *Proceedings of the 1982 Mid-America Linguistics Conference*, ed. Frances Ingemann, 3–29. Lawrence, KS: University of Kansas.
Kiparsky, Paul. 1985. Some consequences of Lexical Phonology. *Phonology Yearbook* 2:82–138.
Koopman, Hilda, and Anna Szabolcsi. 2000. *Verbal complexes*. Cambridge, MA: The MIT Press.

Kratzer, Angelika, and Elisabeth Selkirk. 2007. Phase theory and prosodic spellout: The case of verbs. *The Linguistic Review* 24:93–135.

Krivokapić, Jelena. 2007. Prosodic planning: Effects of phrasal length and complexity on pause duration. *Journal of Phonetics* 35:162–179.

Levelt, Willem J. M. 1989. *Speaking: From intention to articulation*. Cambridge, Ma.: MIT Press.

Levelt, Willem J. M., Ardi Roelofs, and Antje S Meyer. 1999. A theory of lexical access in speech production. *Behavioral and Brain Sciences* 22:1–38.

Liberman, Mark, and Alan Prince. 1977. On stress and linguistic rhythm. *Linguistic Inquiry* 8:249–336.

Lieber, Rochelle. 1980. On the organization of the lexicon. Doctoral Dissertation, MIT.

Marvin, Tatjana. 2002. Topics in the stress and syntax of words. Doctoral Dissertation, MIT.

Mascaró, Joan. 1976. Catalan phonology and the phonological cycle. Doctoral Dissertation, MIT.

McGregor, William B. 2011. *The Nyulnyul language of Dampier Land, Western Australia*. Canberra: Australian National University.

Miller, George A. 1956. The magical number seven, plus or minus two: some limits on our capacity for processing information. *The Psychological Review* 63:81–97.

Neeleman, Ad, and Hans van de Koot. 2006. On syntactic and phonological representations. *Lingua* 116:1524–1552.

Nespor, Marina, and Irene Vogel. 1986. *Prosodic Phonology*. Dordrecht: Foris.

Newell, Heather. 2008. Aspects of the morphology and phonology of phases. Doctoral Dissertation, McGill University, Montréal.

Nhàn, Ngô Thanh. 1984. The syllabeme and patterns of word formation in Vietnamese. Doctoral Dissertation, New York University, New York, NY.

Ó Siadhail, Mícheál. 1988. *Learning Irish*. New Haven, CT: Yale University Press.

Pentland, Christina, and Mary Laughren. 2004. Distinguishing prosodic word and phonological word in Warlpiri: prosodic constituency in morphologically complex words. In *Proceedings of the 2004 Conference of the Australian Linguistic Society*.

Peperkamp, Sharon. 1997. *Prosodic words*. The Hague: Holland Academic Graphics.

Pesetsky, David. 1979. Russian morphology and lexical theory. Ms., MIT.

Pham, Andrea. 2008. Is there a prosodic word in Vietnamese? *Toronto Working Papers in Linguistics* 29:1–23.

Pinker, Steven, and Ray Jackendoff. 2005a. The faculty of language: what's special about it? *Cognition* 95:201–236.

Pinker, Steven, and Ray Jackendoff. 2005b. The nature of the language faculty and its

implications for the evolution of language (Reply to Fitch, Hauser and Chomsky). *Cognition* 97:211–225.
Prince, Alan S. 1980. A metrical theory for Estonian quantity. *Linguistic Inquiry* 11:511–562.
Prince, Alan S., and Paul Smolensky. 2004. *Optimality Theory: Constraint Interaction in Generative Grammar*. Malden, MA: Blackwell. Revised and annotated edition. Previously published as Rutgers University Center for Cognitive Science Tech report #2.
Revithiadou, Anthi. 2011. The phonological word. In *The Blackwell companion to phonology. Volume IV: Phonological interfaces*. Wiley-Blackwell.
Rubach, Jerzy. 1985. Abstract vowels in three dimensional phonology: The yers. *The Linguistic Review* 5:247–280.
Sammallahti, Pekka. 1998. *The Saami languages: an introduction*. Karasjok: Davvi Girji OS.
Schapper, Antoinette. 2022. *A grammar of Bunaq*. Amsterdam: Mouton de Gruyter.
Scheer, Tobias. 2011. *A guide to morphosyntax-phonology interface theories*. Berlin: Mouton de Gruyter.
Scheer, Tobias. 2012. *Direct interface and one-channel translation*. Berlin: Walter de Gruyter.
Schiering, René, Balthasar Bickel, and Kristine A. Hildebrandt. 2010. The prosodic word is not universal, but emergent. *Journal of Linguistics* 46:657–709.
Selkirk, Elisabeth. 1972. The phrase phonology of English and French. Doctoral Dissertation, MIT.
Selkirk, Elisabeth. 1980. Prosodic domains in phonology: Sanskrit revisited. In *Juncture*, ed. Mark Aronoff and Mary-Louise Keans, 107–129. Saratoga, CA: Anma Libri.
Selkirk, Elisabeth. 1981. On prosodic structure and its relation to syntactic structure. In *Nordic Prosody II*, ed. Thorstein Fretheim, 111–140. Trondheim: Tapir.
Selkirk, Elisabeth. 1982. *The syntax of words*. Cambridge, MA: The MIT Press.
Selkirk, Elisabeth. 1984. *Phonology and syntax: the relation between sound and structure*. Cambridge, MA: MIT Press.
Selkirk, Elisabeth. 1986. On derived domains in sentence phonology. *Phonology Yearbook* 3:371–405.
Selkirk, Elisabeth. 1996. The prosodic structure of function words. In *Signal to syntax*, ed. James L. Morgan and Katherine Demuth, 187–213. Mahwah, NJ: Lawrence Erlbaum.
Selkirk, Elisabeth. 2000. The interaction of constraints on prosodic phrasing. In *Prosody, theory and experiment: Studies presented to Gösta Bruce*, 231–261. Dordrecht:

Kluwer.

Selkirk, Elisabeth. 2011. The syntax-phonology interface. In *The Handbook of Phonological Theory*, ed. John Goldsmith, Jason Riggle, and Alan C. L. Yu, 435–484. Wiley-Blackwell. Revised and updated 2nd edition.

Shallice, Tim, Raffaella I. Rumiati, and Antonella Zadini. 2000. The selective impairment of the phonological output buffer. *Cognitive Neuropsychology* 17:517–546.

Shwayder, Kobey. 2015. Words and subwords: phonology in a piece-based syntactic morphology. Doctoral Dissertation, University of Pennsylvania.

Sproat, Richard. 1985. On deriving the lexicon. Doctoral Dissertation, MIT.

Stump, Gregory T. 2001. *Inflectional Morphology. A Theory of Paradigm Structure*. Cambridge: Cambridge University Press.

Svenonius, Peter. 2016a. Complex predicates as complementation structures. In *Approaches to complex predicates*, ed. Léa Nash and Pollet Samvelian, 212–247. Leiden: Koninklijke Brill.

Svenonius, Peter. 2016b. Spans and words. In *Morphological metatheory*, ed. Daniel Siddiqi and Heidi Harley, 201–222. Amsterdam: John Benjamins.

Tallman, Adam J. R. 2020. Beyond grammatical and phonological words. *Language and Linguistics Compass* 14:e12364.

Thompson, Laurence C. 1965. *A Vietnamese reference grammar*. Honolulu, HI: University of Hawaii Press.

Travis, Lisa. 1984. Parameters and effects of word order variation. Doctoral Dissertation, Massachusetts Institute of Technology, Cambridge, Ma.

Trips, Carola, and Jaklin Kornfilt. 2017. Further insights into phrasal compounding. In *Further investigations into the nature of phrasal compounding*, ed. Carola Trips and Jaklin Kornfilt, 1–11. Berlin: Language Science Press.

Vigario, Marina. 2010. Prosodic structure between the prosodic word and the phonological phrase: recursive nodes or an independent domain? *The Linguistic Review* 27:485–530.

Vogel, Irene. 2009. The status of the clitc group. In *Phonological domains: universals and deviations*, ed. Janet Grijzenhout and Barış Kabak, 15–46. Berlin: Mouton de Gruyter.

Vogel, Irene. 2020. Life after the Strict Layer Hypothesis: prosodic structure geometry. In *Prosodic studies: challenges and prospects*, ed. Hongming Zhang and Youyong Qian, 9–60. London: Routledge.

Wangdi, Pema. 2021. What is a word in Brokpa? *Linguistics of the Tibeto-Burman Area* 44:264–296.

Wheeldon, Linda R, and Aditi Lahiri. 2002. The minimal unit of phonological encoding: Prosodic or lexical word. *Cognition* 85:B31–B41.

Wheeler, Max W. 2005. *The phonology of Catalan*. Oxford: Oxford University Press.

Whitney, William Dwight. 1889. *Sanskrit grammar, including both the classical language and the older dialects, of Veda and Brahmana*. Cambridge, MA: Harvard University Press, 2nd edition edition.

Williams, Edwin. 2007. Dumping lexicalism. In *The Oxford handbook of linguistic interfaces*, ed. Gillian Ramchand and Charles Reiss, 353–381. Oxford University Press.

Woodbury, Anthony C. 1987. Meaningful phonological processes: a consideration of Central Alaskan Yupik eskimo prosody. *Language* 63:685–740.

Woodbury, Anthony C. 2024. Constituency in Cup'ik and the problem of holophrasis. In *Constituency and convergence in the Americas*, ed. Adam J. R. Tallman, Sandra Auderset, and Hiroto Uchihara, 83–136. Berlin: Language Science Press.

Wynne, Hilary S. Z., Linda Wheeldon, and Aditi Lahiri. 2018. Compounds, phrases and clitics in connected speech. *Journal of Memory and Language* 98:45–58.

Zwicky, Arnold M. 1977. *On clitics*. Bloomington, IN: Indiana University Linguistics Club.

Zwicky, Arnold M. 1987. Suppressing the Zs. *Journal of Linguistics* 23:133–148.